

SILAS M. FALDE

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EDUCATION

University of Michigan School of Engineering
Senior B.E.S. in Data Science, GPA: 3.09/4.00

Ann Arbor, MI
August 2023 – December 2026

Relevant Courses: Machine Learning, Artificial Intelligence, Computer Vision, Computational Modelling of Complex Systems, Database Management, Regression Analysis, Probability and Statistics, Bayesian Data Analysis, Data Structures and Algorithms

EXPERIENCE

Ally Financial, Inc.

Intern, Artificial Intelligence & Advanced Analytics

Detroit, MI
May 2026 – August 2026

- Prototyped large language model workflows to automate internal tasks and improve efficiency and safety.
- Built Python NLP pipelines to evaluate PII masking and redaction model performance for sensitive company documents.
- Enhanced a data dictionary migration tool by adding batch processing, smarter metadata selection, and a streamlined interface to improve usability and efficiency.

University of Michigan Department of Epidemiology

Research Assistant – Computational Lead

Ann Arbor, MI
September 2025 – May 2026

- Led development of a Python pipeline to classify collective-violence narratives using prompt-based large language model inference and structured response processing.
- Built reusable preprocessing utilities for NVDRS-style text, demographic derivation, path management, and annotation-round data handling.
- Designed human annotation workflows and measured inter-annotator agreement with Cohen's kappa and Krippendorff's alpha to resolve disagreements and assess model performance.

Ally Financial, Inc.

Intern, Data Management

Detroit, MI
May 2025 – August 2025

- Created data dictionaries and lineage documentation in Python to improve data governance and compliance.
- Added data quality checks to existing data pipelines with Python and SQL to improve data integrity and reliability.
- Developed a Power BI data availability dashboard to monitor refresh rates and availability for critical data assets.

University of Michigan Department of Epidemiology

Research Assistant

Ann Arbor, MI
September 2024 – May 2025

- Earned 3rd place in the CDC Youth Mental Health: Novel Variables Data Challenge, placing in roughly the top 1% of entries.
- Developed Python pipelines on a high-performance computing cluster to clean, normalize, and engineer features from more than 400,000 suicide-victim narratives.
- Used large language models to extract structured trends from narrative text and evaluated model-assisted outputs with classification metrics and error analysis.
- Documented findings with reproducible analyses, reports, and visualizations for the research team and publication.

PROJECTS

Home Server Infrastructure

Systems Developer

Ann Arbor, MI
2024 – Present

- Designed and operate a single-host Ubuntu self-hosting platform using Docker Compose stacks for ingress, media, cloud storage, AI services, monitoring, automation, and game hosting.
- Automated media workflows with Bash and Python for queueing, GPU-accelerated HEVC encoding, staging, metadata-safe publishing, library audits, and failure-aware batch processing.
- Configured Cloudflare Tunnel and Caddy ingress, Tailscale private administration, systemd scheduling, service health checks, and documented upgrade, backup, recovery, and security procedures.

FPS PvP Dynamics

Project Developer

Ann Arbor, MI
March 2026 – April 2026

- Built a configurable **agent-based simulation** in Python to model player movement, combat, objectives, respawning, and stochastic per-tick match dynamics.
- Developed trace capture, metrics export, visualization, and automated tests for debugging and validating simulation behavior.
- Ran parallel parameter sweeps, replication studies, **Latin hypercube sampling**, and **Sobol sensitivity analysis** to quantify the effect of model parameters.

Photo Processing Helper

Project Developer

Ann Arbor, MI
Jan 2026 - Present

- Built a modular Python CLI and library for photo framing, collage generation, panorama stitching, and Nikon RAW-to-JPG matching across large image folders.
- Implemented OpenCV, NumPy, Pillow, and rawpy pipelines for resizing, cropping, alignment, blending, and exact-dimension output while preserving high-bit-depth detail.
- Added tests and validation for file discovery, output integrity, stitching behavior, and repeatable batch processing.

Predicting Survival in the ICU

Project Developer

Ann Arbor, MI
January 2025 – March 2025

- Developed logistic-regression, convolutional, and recurrent-neural-network models in Scikit-learn and PyTorch to predict ICU mortality from clinical time-series and static features.
- Built reproducible data loaders and preprocessing for missing values, temporal observations, and engineered clinical features, with configuration-driven training and evaluation.
- Evaluated models with AUROC, visualization, and automated tests covering CNN, transformer, and output behavior.

SKILLS

Programming: Python, C++, Bash, JavaScript, R, SQL

Machine Learning: PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, OpenCV, spaCy, NLTK

Web: HTML, CSS, JavaScript, responsive design, static-site publishing

Systems: Linux, Docker Compose, systemd, CMake, Make, Git, CI/CD

Software: VS Code, Microsoft Office Suite, Google Workspace, Snowflake, Collibra, Alation, Manta